A First Course in Abstract Algebra

Section 33: Finite Fields

Detailed worked solutions

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Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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33 Finite Fields

Exercise 33.1: A power of two

The factorization is $4096 = 2^{12}$. For every prime p and positive integer n, there exists a finite field of order p^n. Thus a field with 4096 elements exists, namely $\mathrm{GF}(2^{12})$.

Exercise 33.2: An integer with two distinct prime factors

We have $3127 = 53 \cdot 59$. Both factors are prime and distinct, so this is not a prime power. Every finite field has prime-power order; therefore no field of 3127 elements exists.

Exercise 33.3: A cube of a prime

Compute $41^2 = 1681$ and $1681 \cdot 41 = 68{,}921$. Since 41 is prime, $68{,}921 = 41^3$ is a prime power, so $\mathrm{GF}(41^3)$ exists and has the required number of elements.

Exercise 33.4: Primitive eighth roots

The nonzero elements of GF(9) form a cyclic group of order 8. A primitive eighth root is exactly a generator of this group. A cyclic group of order eight has $\varphi(8) = 4$ generators, corresponding to powers with exponents $1, 3, 5, 7$ relatively prime to eight. Thus the answer is four.

Exercise 33.5: Primitive eighteenth roots

The multiplicative group of GF(19) is cyclic of order 18. Its elements of order 18 are its $\varphi(18) = 18(1 - 1/2)(1 - 1/3) = 6$ generators. Hence there are six primitive eighteenth roots of unity.

Exercise 33.6: Primitive fifteenth roots

The multiplicative group of GF(31) is cyclic of order 30. It has a unique subgroup of order 15, whose generators are exactly the elements of order 15 in the whole group. Their number is $\varphi(15) = 15(1 - 1/3)(1 - 1/5) = 8$. Thus the answer is eight.

Exercise 33.7: An order absent from the multiplicative group

The nonzero elements of GF(23) form a group of order 22. An element's order must divide 22, but 10 does not. Therefore there are no elements of order 10, and hence zero primitive tenth roots of unity. Merely computing $\varphi(10)$ without checking divisibility would give an incorrect answer.

Exercise 33.8: True and false statements

(a) True. The multiplicative group of a finite field is cyclic.

(b) False. The additive group of $\mathrm{GF}(p^n)$ is $(\mathbb{Z}_p)^n$; for $n > 1$ every nonzero element has order p, so it is not cyclic.

(c) True. The complex zeros of $x^{28} - 1$ are $e^{2\pi ik/28}$ and form the cyclic group generated by $e^{2\pi i/28}$.

(d) False. $60 = 2^2 \cdot 3 \cdot 5$ is not a prime power.

(e) True. $125 = 5^3$ is a prime power.

(f) False. $36 = 2^2 \cdot 3^2$ is not a prime power.

(g) True. The powers $i, i^2 = -1, i^3 = -i, i^4 = 1$ show that i has order four.

(h) True. Over every finite field there is an irreducible polynomial of each positive degree, including degree 58 over $\mathbb{Z}_2$.

(i) False. Any proposed rational generator has only finitely many primes in its numerator and denominator; its integer powers cannot produce a prime absent from that list. Thus $\mathbb{Q}^*$ is not cyclic.

(j) True. An isomorphic image has the same finite size as F, and within the fixed algebraic closure there is exactly one subfield of that size, the roots of $x^{|F|} - x$. The image is therefore F itself.

Exercise 33.9: Two cubic generators inside one algebraic closure

Both binary cubics $x^3 + x^2 + 1$ and $x^3 + x + 1$ have value 1 at $0, 1$, so both are irreducible. Thus $\mathbb{Z}_2(\alpha)$ and $\mathbb{Z}_2(\beta)$ each have eight elements. Every element of either field satisfies $x^8 - x = 0$. In the common algebraic closure this polynomial has at most eight roots, and each of these fields already consists of eight roots. They must therefore be the same set, proving equality of fields rather than just an abstract isomorphism.

Exercise 33.10: Irreducibles divide a Frobenius polynomial

Let $f \in \mathbb{Z}_p[x]$ be irreducible of degree n, replacing it by a monic associate if necessary. The quotient $E = \mathbb{Z}_p[x]/(f)$ is a field of p^n elements. For $\alpha = x + (f)$, Lagrange's theorem in E^* (and the zero case) gives $\alpha^{p^n} = \alpha$. The minimal polynomial of α is the monic associate of f, so it divides every polynomial vanishing at α, in particular $x^{p^n} - x$. Hence f divides that polynomial up to its harmless unit coefficient.

Exercise 33.11: A multiplicative generator has full degree

Every nonzero element of F is a power of α, so belongs to the subfield $\mathbb{Z}_p(\alpha)$. Zero also belongs, hence $\mathbb{Z}_p(\alpha) = F$. If $d = \deg(\alpha, \mathbb{Z}_p)$, the simple extension has p^d elements. Comparing with $|F| = p^n$ gives $d = n$.

Exercise 33.12: Subfields of a finite field

Let $E = \mathrm{GF}(p^n)$ and $m \mid n$. Set $K = \{a \in E : a^{p^m} = a\}$. Iterated Frobenius gives $(a+b)^{p^m} = a^{p^m} + b^{p^m}$ and $(ab)^{p^m} = a^{p^m} b^{p^m}$, so K is closed under addition, multiplication, and negatives; inverses of nonzero

fixed elements are fixed too. It contains $0, 1$, hence is a subfield. Because $p^m - 1$ divides $p^n - 1$, the cyclic group E^* has exactly $p^m - 1$ solutions of $a^{p^m-1} = 1$. Including zero gives $|K| = p^m$. Any subfield of size p^m consists of roots of $x^{p^m} - x$, so must equal this K by counting. This proves uniqueness. Conversely any subfield has size p^m, and the tower law makes m divide n.

Exercise 33.13: Factoring a Frobenius polynomial

The derivative of $x^{p^n} - x$ is -1, so it has no repeated irreducible factor. An irreducible monic polynomial f of degree d divides it exactly when its roots lie in $\mathrm{GF}(p^n)$. If such a root a lies there, $\mathbb{Z}_p(a)$ is a subfield of order p^d, so $d \mid n$ by the tower law. Conversely if $d \mid n$, the field $\mathbb{Z}_p(a)$ of p^d elements embeds as the unique subfield of $\mathrm{GF}(p^n)$ of that size; all its elements satisfy $x^{p^n} = x$, so f divides the polynomial. Thus its monic irreducible factors are exactly those whose degrees divide n, each appearing once. Since the polynomial is monic, the product has no additional nonunit constant.

Exercise 33.14: Euler criterion and a quadratic over seventeen elements

(a) Choose a generator g of $\mathbb{Z}_p^*$, which has even order $p - 1$. Write the nonzero residue $a = g^k$. It is a square exactly when k is even: the squares are the even powers of g. Also

$$a^{(p-1)/2} = 1 \quad \Longleftrightarrow \quad (p-1) \mid k(p-1)/2 \quad \Longleftrightarrow \quad 2 \mid k.$$

This proves the criterion, including both implications. **(b)** Modulo seventeen, $6^2 \equiv 2$, $6^4 \equiv 4$, and $6^8 \equiv 16 = -1 \neq 1$. Hence six is not a quadratic residue, so $x^2 - 6$ has no root in $\mathbb{Z}_{17}$ and is irreducible there.